Low-Dimensional, Non-Lexical Trajectories of Evolving Software Objects

Giovanni Pezzin
Physis Research Program

September 28, 2026

Abstract

Revisions of a single source file occupy a low-dimensional, temporally directed region of embedding space — but the identity half of the usual trajectory story does not survive a trivial baseline. On an external replication of 434 source files and 11,312 revisions across ten public repositories in five languages, the participation ratio is 2.13 and the effective rank 3.06: 7–10× more concentrated than an isotropic cloud of the same shape, against a per-metric null. PC1+PC2 carry 0.843 of the variance against a null of 0.113. Direction is present and is *not* text overlap: the geometric drift statistic anti-correlates with the lexical one ($\rho = -0.593$) and keeps a partial correlation with revision depth of 0.232 after controlling for lexical adjacency. Identity, however, is not evidence for a semantic representation — same-file revisions separate from cross-file revisions at 0.99 under nomic, but the trivial lexical baselines do the same (token-jaccard 0.988, edit distance 0.974) and separate them *more* by gap median (0.369 and 0.516 against 0.214). We therefore keep the dimensional and directional claims, drop identity-as-semantic-evidence, and make no operational prediction claim: that experiment was never run. Earlier internal figures on six files (2–3 dimensions from 57–82% raw PC1+PC2 variance, no null) are superseded — raw explained variance cannot support a dimensionality claim, as the replication's first control demonstrated.

1 Introduction

Software objects evolve. A source file accumulates revisions; each revision is a state of the object at a point in time. The question is whether these states form a *trajectory* — a continuous, directed path through representation space — or a set of independent points.

This question has practical implications. If revisions form a trajectory, then (a) semantic search should use trajectory queries, not point queries; (b) change prediction should extrapolate the trajectory, not predict from a static embedding; (c) object identity should be defined by trajectory proximity, not point proximity.

We test this on six files from a Rust codebase, using BGE-base embeddings (768 dimensions, `onnx-explicit` backend). We measure three properties: identity (within-object distance vs. between-object distance), direction (Spearman correlation of drift with time, against a shuffled-time null), and structure (spectral concentration of the trajectory).

2 Method

For each file f, we:

1. Extract all revisions $r_1, \ldots, r_n$ from git history.
2. Embed each revision (chunked at 1500 characters, mean-pooled into a single 768-d vector).
3. Compute drift $d_i = \|v_i - v_1\|$ (distance from the first revision).
4. Compute Spearman correlation $\rho(i, d_i)$ against a 2000-permutation shuffled-time null.
5. Compute spectral concentration: the fraction of variance in the first two eigenvalues of the centered Gram matrix.

3 Data

Six files from a Rust codebase (`physis-pro`), spanning 8–27 revisions each:

File	Revisions	Drift range	Spearman ρ	2D var
`hypothesis.rs`	12	0.026	+0.930	0.766
`main.rs`	27	0.116	+0.626	0.621
`studio.rs`	17	0.037	+0.939	0.761
`act.rs`	8	0.053	+0.976	0.817
`ontology.rs`	8	0.043	+0.929	0.740
`quality.rs`	9	0.025	+0.250	0.569

Table 1: Trajectory analysis of six source files. Drift range is the difference between maximum and minimum distance from the first revision. 2D var is the fraction of variance explained by the first two eigenvalues of the Gram matrix.

4 Results

4.1 Identity

Within-object distances (mean pairwise cosine distance between revisions of the same file) are 4–14× smaller than between-object distances (mean cosine distance between revisions of different files). A file's revisions form a tight region of the embedding field; a different file is elsewhere.

4.2 Direction

Five of six files show temporal direction: Spearman ρ ranges from +0.626 to +0.976, all outside the 2000-permutation shuffled-time null (which spans approximately [-0.4, +0.7] for most files). The sixth file, `quality.rs`, fails: $\rho = +0.250$, inside the null. Its drift profile shows why: it wanders and comes back, so distance-from-origin stops being a clock.

4.3 Structure

The first two eigenvalues of the centered Gram matrix carry 0.569–0.817 of the variance, and the files occupy few effective dimensions. This figure alone does *not* establish low intrinsic dimension: a random walk in 768 dimensions also concentrates variance, and an explained-variance fraction is invariant under permuting the revisions, so a row-shuffled null reproduces it exactly. Section 5 reports the statistic against a null that can actually fail.

4.4 Granularity

At document level, identity survives: the document's pooled embedding is closer to its own sections than to another document's sections (nesting margin 0.02–0.03). At paragraph level, identity dissolves: a paragraph of one document is nearly as far from another paragraph of the same document (0.379) as from a paragraph of a different document (0.405).

Positional direction — whether a text moves away from its opening as it proceeds — is a property of genre, not of text. A developmental document (the world-map machine plan) shows strong positional direction at section level ($\rho = +0.665$); a reference document (WHAT_PHYSIS_IS) does not ($\rho = +0.083$).

5 External replication

The six-file result was a single repository and one embedder. We replicated the same three tests on ten public repositories (ripgrep, serde, requests, flask and others; C, C++, JS, Python, Rust): 434 source files, 11,312 revisions. Representation is nomic-embed-text (768-d) via a local ollama server — BGE was unavailable on the replication host, so the embedder changed as well as the corpus. Inference unit is the file. Result: `research/trajectories_external/p2_results.json`; generator `research/papers/p2_analysis.py --task all`.

5.1 Dimension: holds

Against an isotropic cloud of the same shape $n \times d$, all three statistics say the real spectra are far more concentrated: PC1+PC2 share 0.843 vs 0.113 (7.5×), participation ratio 2.127 vs 22.362 (10.5×), effective rank 3.063 vs 22.674 (7.4×). The participation ratio of 2.13 and effective rank of 3.06 are what "2–3 dimensional" should have meant in Section 1; the direction of the comparison differs by statistic, since a *low* participation ratio is what concentration means.

5.2 Direction: holds, and is not lexical

Quantity	Value
corr(geometric adjacency, lexical adjacency)	+0.826
corr(geometric trend, lexical trend)	−0.593
partial corr(geometric adjacency, n_{rev} | lexical adjacency)	+0.232
median geometric adjacency	0.993
median lexical adjacency	0.941

The geometric temporal statistic does not merely survive removing the text: it points the opposite way to the lexical trend (−0.593) and keeps a partial correlation with revision depth of 0.232. This is the part of the trajectory claim that a text-overlap baseline cannot reproduce.

5.3 Identity: does not hold as semantic evidence

Representation	frac. same closer	CI95	gap median
nomic (768-d)	0.9906	[0.976, 0.996]	0.214
token-jaccard (lexical)	0.9883	[0.973, 0.995]	0.369
edit distance (lexical)	0.9742	[0.954, 0.986]	0.516

All three separate same-object from cross-object revisions at 97–99%, and on the gap medians the trivial baselines are *better* than the embedding. Identity across revisions is real; it is not evidence for an embedding. The six-file version of this claim should be read the same way.

5.4 Prediction: not claimed

No next-step extrapolation test exists in the replication artifact (its keys are identity, overlap, dimensionality). We do not claim operational trajectory prediction, here or anywhere.

6 Discussion

What survives, and what does not. Revisions of a single file do form a low-dimensional, temporally directed region: the dimension claim holds against a null that can fail, and the direction statistic is not reproducible by text overlap. But the third pillar of the original framing — that identity is evidence for a semantic representation — does not survive, and the operational prediction that the introduction hoped for was never tested. The honest claim is narrower than "software objects are trajectories": the trajectory is a real geometric property of the revision sequence, and what it is *for* remains open.

The failure mode is informative. `quality.rs` fails the direction test because it wanders and returns. This is the honest failure mode of trajectory analysis: when an object's evolution is not monotonic, distance-from-origin is not a clock. The failure is visible in the same table as the successes, not averaged away.

Identity dissolves below document granularity. A document's paragraphs are not a tighter cluster than paragraphs from different documents. Whatever "the same object" means at revision scale, it is not recoverable from a paragraph. This is consistent with the E80 finding that pooling a project into one vector destroys signal.

Limits. One repository, one embedder (BGE-base-en-v1.5), six files, one metadata axis (commit time). The trajectory instrument is available as `physis-core/examples/reference_spectrum.rs`; the harness is `benchmarks/conatus/e78_reference_spectrum.py`. The The expansion to thousands of files across public repos that this paragraph previously named as the highest-priority replication has now been run (Section 5): 434 files, 11,312 revisions, ten repos. It is the source of the abstract's numbers, and it is what narrowed this paper.